Presentation Mode — Stepper & Gutter System

Handoff document · approved design, full-page renders, implementation spec · August 15, 2026

Summary

Presentation mode's biggest repeated visual flaw was fixed as a system, approved by the owner from pixel-faithful live-render mocks. The – + stepper pair — previously orphaned in the right margin, outside the content column — now attaches to the value it edits, while computed rows keep their digits on the same shared boundary with an empty gutter. The rule strengthens the product's goal: steppers invite readers to play with the inputs; the quiet gutter next to a result says "this one answers back."

The rule (approved)

All digits share one right boundary; the gutter to its right belongs to interaction.

1. Every standard row right-aligns its number on a single shared boundary.
2. A fixed gutter (stepper width + 6 px ≈ 64 px) sits right of that boundary.
3. Input rows fill the gutter with – + , 6 px off the number. With inline units the order is "8.25 % – +" — unit glued to digits, steppers last.
4. Computed rows leave the gutter empty; no – + plus ink color (vs. accent + dashed underline) marks a result.
5. Calcs with no steppers anywhere (all-slider calcs like mortgage-calculator) reserve nothing and render unchanged.

Design-language precedent

Airbnb's guest picker and Stripe's quantity control treat count + buttons as one object; native number fields and macOS steppers put spinners at the field's right edge; iOS HIG says place a stepper adjacent to the value it modifies. Nobody floats controls at the page margin. The trailing "value – +" order (rather than "– value +") is the adaptation that preserves the worked-document's right-aligned digit column. Financial-calculator sites dodge the input/result question with separate panels; instacalc interleaves them, so the within-column cues carry the meaning.

The defect

On every editable numeric row of every stepper calc, .stepper-pair rendered as its own far-right slot outside the content column: at a 1280 px viewport the card's text column ends at x = 1012 while the steppers sat at x = 1023–1081 — ~90 px of empty paper between the number and its controls. It appeared on the exam's a/b/c rows and throughout the kitchen sink (sections 4–7, 10, 15, 19–20).

BEFORE — production render (Algebra I exam, Problem 6 coefficients)

AFTER — approved: steppers attach to the value; digits keep one boundary

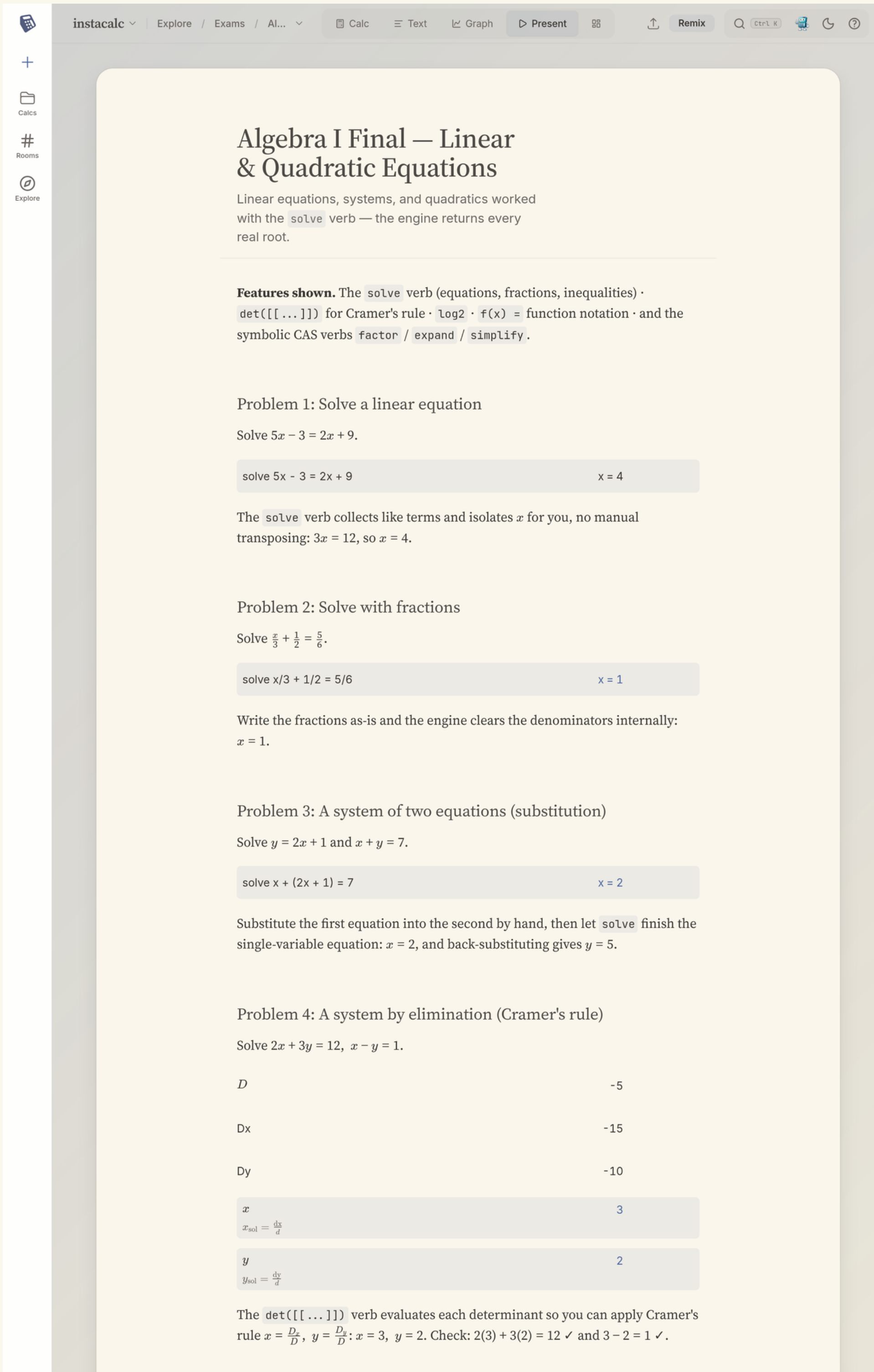
Both images are the real product render; the "after" relocates only the stepper node. Fonts, colors, underline, spacing: untouched.

Detail: inline units stay glued to the digits

"8.25 % – +" and "95 % – +" — value, unit, steppers. Unit-below-value rows ("180" over "lbs") are unchanged.

Full-calc render — Algebra I exam (flagship content)

Complete /present page with the system applied. Note Problem 6: inputs “a 2 – +”, “b 3 – +”, “c –5 – +” stack directly above computed “discriminant 49”, all digits on one right edge.



instacalc | Explore / Exams / Al... | Calc | Text | Graph | Present | Remix | Ctrl K

Calcs
Rooms
Explore

Algebra I Final — Linear & Quadratic Equations

Linear equations, systems, and quadratics worked with the `solve` verb — the engine returns every real root.

Features shown. The `solve` verb (equations, fractions, inequalities) · `det([[...]])` for Cramer's rule · `log2` · `f(x) =` function notation · and the symbolic CAS verbs `factor` / `expand` / `simplify`.

Problem 1: Solve a linear equation

Solve $5x - 3 = 2x + 9$.

solve 5x - 3 = 2x + 9	x = 4

The `solve` verb collects like terms and isolates x for you, no manual transposing: $3x = 12$, so $x = 4$.

Problem 2: Solve with fractions

Solve $\frac{x}{3} + \frac{1}{2} = \frac{5}{6}$.

solve x/3 + 1/2 = 5/6	x = 1

Write the fractions as-is and the engine clears the denominators internally: $x = 1$.

Problem 3: A system of two equations (substitution)

Solve $y = 2x + 1$ and $x + y = 7$.

solve x + (2x + 1) = 7	x = 2

Substitute the first equation into the second by hand, then let `solve` finish the single-variable equation: $x = 2$, and back-substituting gives $y = 5$.

Problem 4: A system by elimination (Cramer's rule)

Solve $2x + 3y = 12,\ x - y = 1$.

D	-5
Dx	-15
Dy	-10
x $x_{\text{sol}} = \frac{\text{dx}}{d}$	3
y $y_{\text{sol}} = \frac{\text{dy}}{d}$	2

The `det([[...]])` verb evaluates each determinant so you can apply Cramer's rule $x = \frac{D_x}{D},\ y = \frac{D_y}{D}$: $x = 3,\ y = 2$. Check: $2(3) + 3(2) = 12$ ✓ and $3 - 2 = 1$ ✓.

exam-algebra1/present — part 1 of 3

Problem 5: Factor-and-solve a quadratic

Solve $x^2 - 5x + 6 = 0$.

```
solve x^2 - 5x + 6 = 0
```
$x^2 - 5x + 6 = 0$ ⟶ $\langle 3,\ 2 \rangle$

`solve` returns *both* roots at once, equivalent to factoring $(x - 2)(x - 3) = 0$: $x = 2$ or $x = 3$.

Problem 6: The quadratic formula, spelled out

Solve $2x^2 + 3x - 5 = 0$ using the quadratic formula.

a	2 − +
b	3 − +
c	-5 − +
discriminant $\text{disc} = b^2 - 4a\,c$	49
$x+$ $x_{\text{plus}} = \frac{-b+\sqrt{\text{disc}}}{2a}$	1
$x-$ $x_{\text{minus}} = \frac{-b-\sqrt{\text{disc}}}{2a}$	-2.5

Name the coefficients, compute the discriminant once, and reuse it in $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Discriminant = 49, so $x = 1$ or $x = -2.5$.

Problem 7: Vertex of a parabola

Find the vertex of $y = 2x^2 + 3x - 5$.

x-vertex $x_{\text{vertex}} = \frac{-b}{2a}$	-0.75
y-vertex $y_{\text{vertex}} = a\,x_{\text{vertex}}^2 + b\,x_{\text{vertex}} + c$	-6.125

The vertex sits at $x = -\frac{b}{2a}$; reusing the named a, b, c from Problem 6 lets you plug it back in for y. Vertex at $(-0.75,\ -6.125)$ — the minimum of this upward parabola.

Problem 8: Solve an inequality

Solve $3x - 4 \le 11$.

```
solve 3x - 4 <= 11
```
x <= 5

`solve` handles inequalities too, reporting the solution set rather than a single value: $x \le 5$.

Problem 9: Exponential — solve for the exponent

Solve $2^x = 32$.

```
log2(32)
```
5

Take a base-2 logarithm to undo the exponent, since $x = \log_2 32$ and `log2` is built in: $x = 5$, since $2^5 = 32$.

Problem 10: Function notation and evaluation

Problem 10: Function notation and evaluation

For $f(x) = x^2 - 3x + 2$, find $f(5)$.

f(5)	12

Instacalc supports $f(x)$ function notation directly, so you define on one row and evaluate on the next: $25 - 15 + 2 = 12$.

Problem 11: Factor a quadratic (symbolic)

Factor $x^2 - 5x + 6$.

factor x^2 - 5x + 6

(x - 2)*(x - 3)

The `factor` verb returns the product of binomials symbolically, not a number: $(x - 2)(x - 3)$ — the same roots Problem 5 solved for, now read off the factors directly.

Problem 12: Expand a binomial (symbolic)

Expand $(x + 3)^2$.

expand (x + 3)^2	x^2 + 6*x + 9

The `expand` verb multiplies it out symbolically, term by term: $x^2 + 6x + 9$ — the perfect-square pattern $x^2 + 2ax + a^2$.

Problem 13: Simplify an expression (symbolic)

Simplify $\frac{x^2-9}{x-3}$.

simplify (x^2 - 9)/(x - 3)	x + 3

The `simplify` verb cancels the common factor $(x - 3)$ symbolically after recognizing $x^2 - 9 = (x - 3)(x + 3)$: $x + 3$ (for $x \neq 3$).

Auto layout

Pre-seeded · review pending

‹ Electric Power Efficiency

Algebra II Final — Functions, Logs & Complex Numbers ›

Want to make this yours?

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Remix this calc →

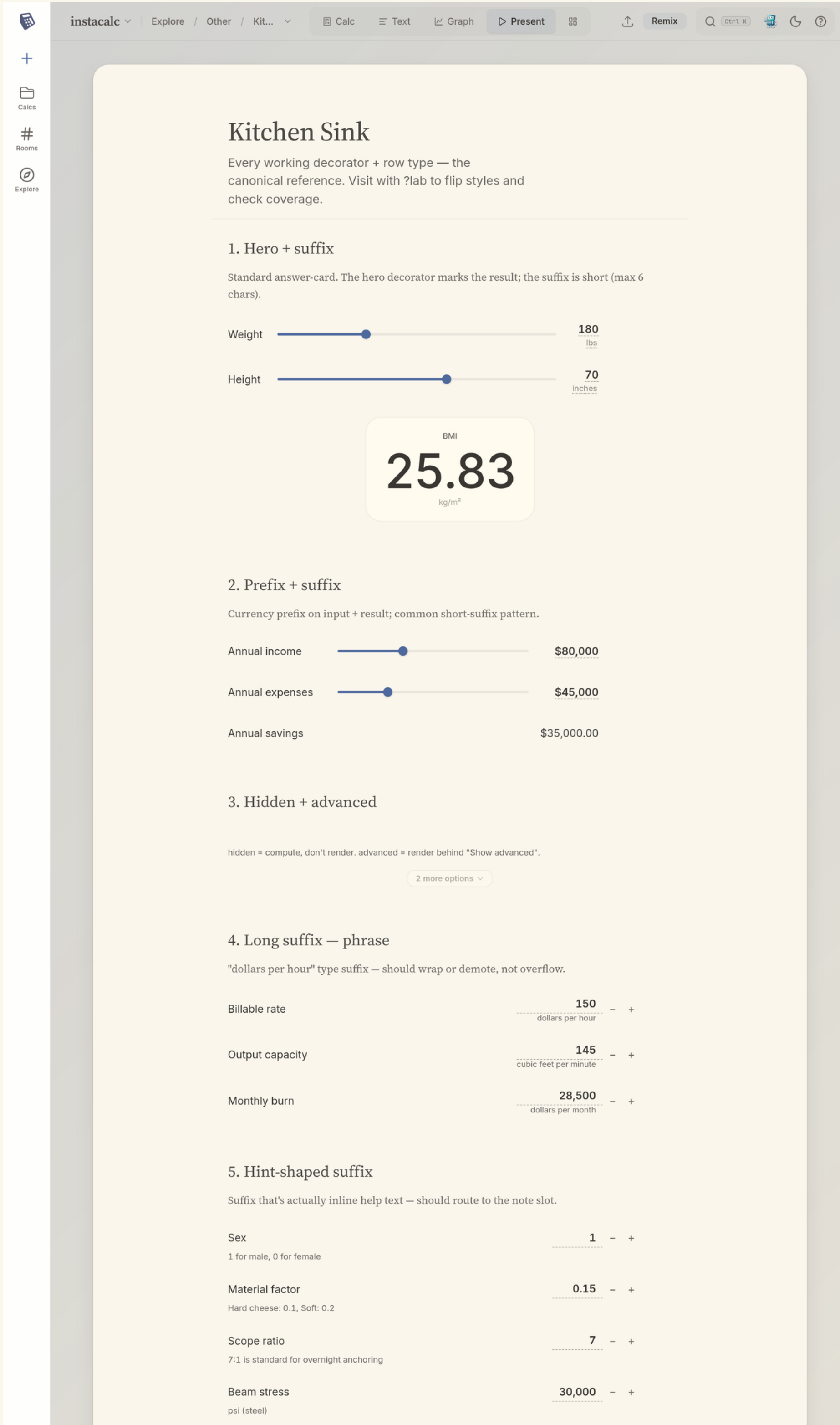
Info

Sign in

exam-algebra1/present — part 3 of 3

Full-calc render — Kitchen Sink (all 27 sections)

The canonical coverage fixture under the same rule: long-suffix rows keep the phrase under the digits with – + beside the number; hint/note/tooltip rows, sections, grid and split layouts, heroes, and readonly rows are unaffected.



__kitchen-sink/present — part 1 of 4

6. Note decorator

Explicit note(...) decorator — same render path as the hint-shaped suffix above.

Shipping cost 9.99 − +
Free over $50

Tax rate 8.25 % − +
Varies by ZIP — verify before checkout

7. Tooltip

tooltip = hover-only help; should not affect layout.

Sample size 30 − +

Confidence 95 % − +

8. Section + nested subsection

Group A — Inputs

Group a input 1 10 − +

Group a input 2 20 − +

Subsection

Nested value 30

9. Collapsed section

Advanced settings

4 rows

10. Accent row

The accent decorator visually highlights the row.

Emergency reserve 50,000 − +

11. Readonly

readonly disables interactivity (cascades from a section marked readonly).

Constant pi 3.142

12. Slider min..max syntax (inline range)

`// min..max by step` is the engine's short slider syntax.

Slider quick 50

Slider with units 250 ml

13. Dropdown

Dropdown — pick from a set of named values.

Plan Hobby

Selected price $9.00

__kitchen-sink/present — part 2 of 4

14. Toggle

Boolean control via .

Premium add-on

Base price 49.99 – +

Total
$49.99

15. on row

Per-row icon decorator.

Fuel 12.5 gal – +

Distance 320 mi – +

16. Calc title + in the first heading at the top of this file

17. Layout: grid (peer outputs)

Color breakdown

RED – 255 +	GREEN – 128 +
BLUE – 64 +	ALPHA – 1 +

18. Layout: split (inputs left, outputs right)

I/O scenario

Revenue
150,000 – +

Expenses
90,000 – +

Profit
60,000

Margin pct
40 %

19. — row spans full grid width

Q4 total revenue across all regions 600,000 – +

20. Format: percent and currency

Percent rate 0.075 – +

Compact dollars 12,345.67 – +

Raw count 1,234,567 – +

20b. Suppress unit on dimensioned results

BMI is technically kg/m^2 but humans read it as a bare number. An empty suffix decorator

__kitchen-sink/present — part 3 of 4

BMI is technically kg/m² but humans read it as a bare number. An empty suffix decorator — suffix() or suffix(none) — opts out of *both* an explicit suffix string AND any unit the engine would derive from dimensional math. Use on indices, ratios, scores.

BMI

25.88

21. Inline variable interpolation in prose

With these inputs the savings reach 35000 per year, or 2917 per month.

22. Narrative row (markdown-friendly)

Narrative rows use **triple-slash**. Supports *markdown*, links, and `inline code`.

23. Plain comment

Plain double-slash comments render dimmed — author notes for readers, not decorators.

24. Divider

Above this line: a horizontal divider drawn from `---` on its own line.

25. Quoted string narrative

Plain string literals (in quotes) also render as prose. Consecutive quoted lines group together with no gap.

26. Comment heading (level 3)

A level-3 sub-label

`### Text` becomes an inline sub-label, smaller than ## section.

27. End cap — one final hero so the calc has a clear payoff

Total saved over horizon

$1,050,000.00

Grid

Pre-seeded · review pending

Account Receivable Days ›

Want to make this yours?

Remix "Kitchen Sink" into your own editable copy — tweak it, share it, make it yours.

Remix this calc →

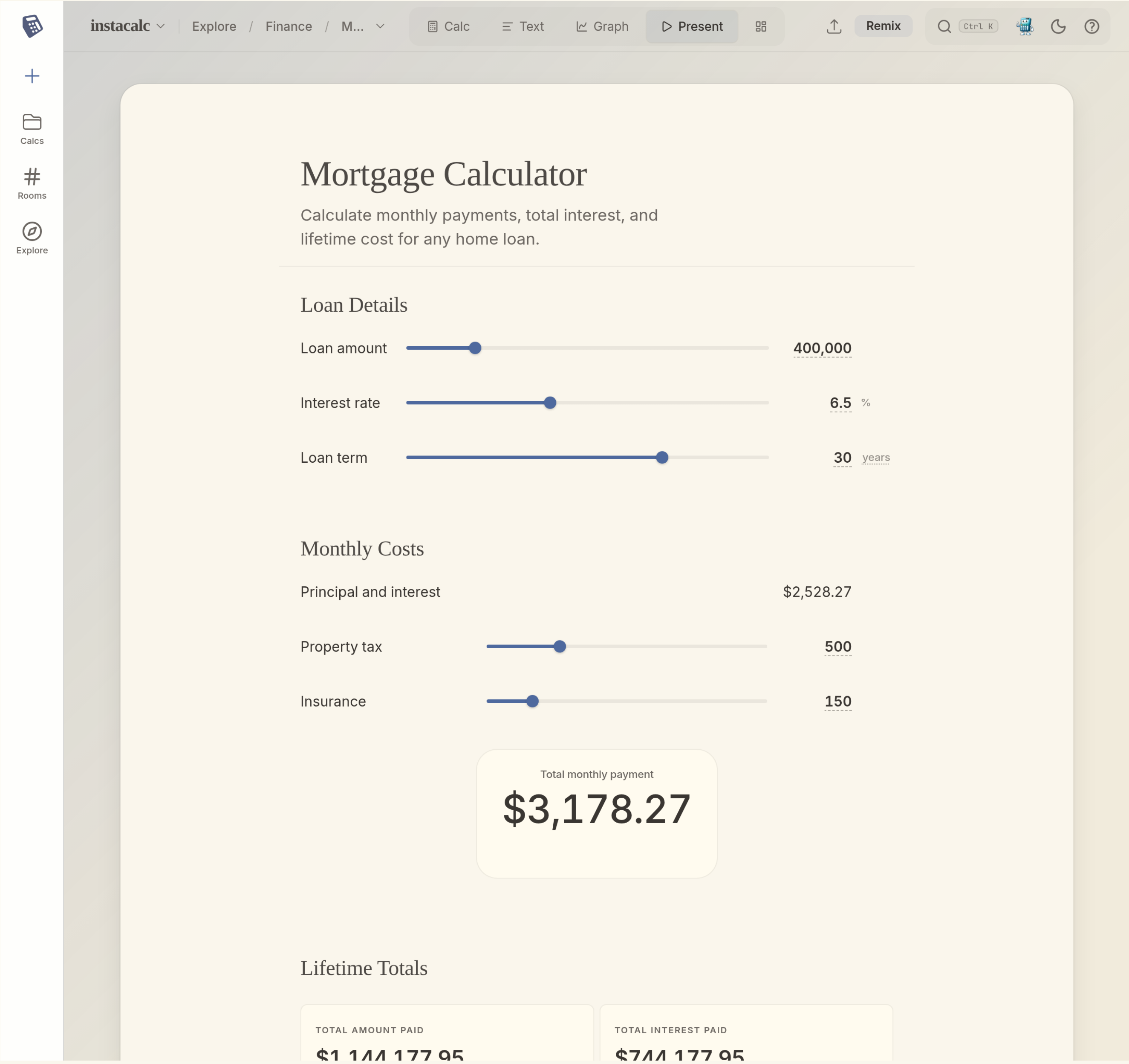
Info

Sign in

__kitchen-sink/present — part 4 of 4

Control case — Mortgage Calculator (unchanged by design)

This calc uses sliders only — no visible steppers — so rule 5 reserves no gutter and the page renders exactly as production. A rule that leaves already-good pages alone is the test that it is a rule, not a patch.



mortgage-calculator/present — production render, no change required

Implementation spec

Target: the presentation row component in kazad/instacalc-private (rendered class hash svelte-1459lf9 on the live site).

• The real fix is the row grid template, not DOM reparenting: the value column ends at the shared boundary and a fixed stepper column (stepper width + 6 px) follows, present whenever the calc has any visible stepper rows.
• Slider rows carry hidden zero-width .stepper-pair nodes — only **visible** steppers count when deciding whether a calc reserves the gutter.
• Inline unit spans currently render as row-level siblings after .row-value; the order must become value, unit, steppers.
• Unit-below-value rows need no change.
• Decide hover behavior explicitly: the approved mocks show steppers always visible.

Reference transform (visual oracle used for the approved renders — not the implementation)

```
const pairs = [...document.querySelectorAll('row .stepper-pair')]
  .filter(sp => sp.getBoundingClientRect().width > 0);
if (pairs.length) {
  const w = Math.max(...pairs.map(sp => sp.getBoundingClientRect().width));
  pairs.forEach(sp => {
    const row = sp.closest('row');
    const rv = row.querySelector('.row-value');
    if (!rv) return;
    sp.style.marginLeft = '6px';
    rv.appendChild(sp);
    const unitWrap = [...row.querySelectorAll('span[class*="text-xs"]')].find(el =>
      !rv.contains(el) && !el.closest('.row-label') && !el.closest('.stepper-pair'));
    if (unitWrap) { unitWrap.style.marginLeft = '2px'; rv.insertBefore(unitWrap, sp); }
  });
  const reserve = w + 6;
  document.querySelectorAll('[data-row-index]').forEach(rowEl => {
    const rv = rowEl.querySelector('row .row-value');
    if (!rv) return;
    const sp = rv.querySelector('.stepper-pair');
    if (!(sp && sp.getBoundingClientRect().width > 0)) rv.style.paddingRight = reserve + 'px';
  });
}
```

Acceptance checks

1. exam-algebra1/present: a/b/c read “2 – +” etc. above “discriminant 49”; digits 2, 3, –5, 49 share one right edge.
2. __kitchen-sink/present: “8.25 % – +”, “95 % – +”; long-suffix rows keep the phrase under the digits; grid/split sections 17–18 unaffected.
3. mortgage-calculator/present: pixel-identical to production.

Status and backlog

Stepper/gutter system	APPROVED. Implementation session spawned from kazad/instacalc-private: “Presentation Mode: stepper/gutter alignment” (session_019nXynad6eMijn6rs9y3QzF), branch presentation-stepper-gutter.

Unit suffix edge cases	Next mock candidate — needs the expanded 65-row __kitchen-sink fixture.
Adjacent @hero rows compete	Not reproducible on live fixtures (single heroes only); needs expanded fixture.
Narrative/note voice inversion	Needs test-present fixture content.
Parser bugs (heading/LaTeX merges, @prefix on negatives, currency decimals)	Code fixes, not visual; do in the instacalc-sourced session.

Working method (carry forward)

The live product is the design language — make small corrections to it, never a new identity. One change at a time; every proposal is a real screenshot beside the identical content with exactly one thing changed; show one example and stop for the owner's reaction. No invented palettes, no monospace numbers, no boxes or chrome the product doesn't have.